LAB 3: ARCHITECTURE SELECTION

Pharmacy Inventory and Automated Reordering System

Name: Jamula Soham Rao

SRN: PES2UG24CS202

Architecture Selection

We chose a Layered Architecture for the Pharmacy Inventory and Automated Reordering System. The design separates the presentation layer, application and business services, and the data and integration layer. The user interface accesses inventory, alert, approval, and security functions only through defined service interfaces.

Two Scenario-Specific Reasons

Clear separation of pharmacy rules. FEFO checkout, expiry thresholds, reorder rules, urgent zero-stock handling, and manager approval are independent business concerns. Placing them in focused components makes each rule easier to test and modify without changing the user interface or database.

A practical fit for a centralized transactional system. The pharmacy requires consistent stock quantities, batch expiry dates, and purchase-order states. A layered design keeps one controlled source of truth while avoiding the deployment, networking, and data-consistency overhead of microservices.

Security Advantage

Security is enforced through a dedicated Security and Access Control component. Role-based permissions distinguish Pharmacy Clerk, Pharmacy Manager, and System Administrator actions. All service traffic uses authenticated TLS 1.2+ connections, supplier and inventory data is protected with AES-256 at rest, and approval or rejection events can be recorded for auditability.

Performance Benefit

In-process service calls reduce latency compared with distributed microservices. Indexed inventory queries and cached menu/reference data can keep FEFO selection and dashboard loading responsive. Expiry scans and reorder generation may run as background jobs, helping the system satisfy the requirement that purchase orders be generated within 10 minutes even during peak load.

Component Model Summary

Components: Pharmacy UI, Security and Access Control, Inventory and FEFO Service, Expiry Alert Service, Reorder and PO Manager, Data Repository, Pharmacy Database, and Supplier Gateway.

Requirement basis: Jira project PROJ, issues PROJ-6 through PROJ-15; duplicate items omitted.